

Private Clinic Obstetrics Management System

User Manual

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1. Introduction

Private Clinic — Obstetrics Management System (OMS) is a comprehensive software application designed for Egyptian maternity hospitals and obstetrics & gynecology clinics. It provides end-to-end management of patient care, from registration through antenatal visits, investigations, delivery, and postpartum follow-up.

The system is bilingual (English and Arabic) with full RTL support, works offline using a local database, and can run from a USB drive with no installation required. It supports multiple facilities (hospitals or clinics) with isolated patient data and independent branding (name, logo, support phone).

Key features include:

- * Patient registration and comprehensive profile management
- * Antenatal visit tracking with vital signs and clinical notes
- * Investigations management (lab results, imaging)
- * ML-based risk classification and assessment
- * Labor & delivery tracking
- * Postpartum care for mother and baby
- * Bilingual interface (English / Arabic)
- * Multi-facility support with independent branding
- * Offline operation — no internet required
- * USB portable — runs without installation

2. Getting Started

2.1 System Requirements

- * Operating System: Windows 10/11, Linux, or macOS
- * RAM: 4 GB minimum, 8 GB recommended
- * Disk Space: 200 MB free
- * Screen Resolution: 1280 x 720 or higher
- * No internet connection required for daily use

2.2 How to Start the Application

Step 1:

Insert the USB drive into your computer.

Step 2:

Open the USB folder and locate the start file:

- * Windows: Double-click start.bat
- * Linux: Double-click start.sh (or run bash start.sh in terminal)
- * macOS: Run bash start.sh in Terminal

Step 3:

Wait for the application to load. Your browser will open automatically.

Step 4:

The system opens at: <http://localhost:5000/dashboard>

NOTE:

If the browser does not open automatically, manually open your browser and navigate to <http://localhost:5000/dashboard>

3. Login & Language Settings

3.1 Default Login Credentials

On first use, log in with the default administrator account:

* Username: admin

* Password: admin123

It is recommended to change the admin password after first login.

3.2 Facility Code

If you are running a single clinic, leave the facility code field empty (DEFAULT will be used automatically). For multi-facility setups, enter your assigned facility code. A facility code is a short identifier like 'CLINIC01' or 'HOSPITAL_A' that links to that facility's data and branding.

You can browse available facilities by clicking the magnifying glass (search) icon next to the facility code field.

3.3 Switching Between English and Arabic

On the login page and throughout the application, you will find a language toggle button. Click it to switch between English and Arabic. The interface instantly updates all text, labels, and messages to the selected language.

3.4 Password Visibility Toggle

When typing your password, click the eye icon (show/hide) next to the password field to reveal or hide the characters. This helps avoid typing errors.

3.5 New User Registration

New users can register by clicking the 'Register' link on the login page. Fill in your name, username, password, and select a role. After registration, an administrator must approve your account before you can log in.

4. Dashboard Overview

After logging in, you will see the main dashboard. This is your central hub for navigating the system. The dashboard displays:

- * Quick statistics: total patients, today's visits, pending investigations
- * Search bar to find patients quickly
- * Navigation sidebar with links to all sections
- * Your facility name and branding at the top
- * Language toggle (English / Arabic)
- * User profile and logout button

Use the sidebar menu to navigate between sections: Patients, Antenatal Care, Investigations, Classification, Delivery, Postpartum, and Admin.

5. Patient Management

5.1 Adding a New Patient

Step 1:

From the sidebar, click 'Patients' then 'Add New Patient'.

Step 2:

Fill in the patient's information:

- * Full name (required)
- * Date of birth
- * Phone number
- * Address
- * Medical history notes

Step 3:

Click 'Save' to register the patient.

The patient will appear in the patient list immediately.

5.2 Searching for Patients

Use the search bar on the dashboard or the Patients page. You can search by:

- * Patient name (partial matches supported)
- * Patient ID
- * Phone number

Results update in real-time as you type.

5.3 Patient Profile

Click on a patient's name to open their profile. The profile page shows all information for that patient, including:

- * Personal details and contact information
- * Medical history
- * Antenatal visit history
- * Investigation results
- * Classification and risk status
- * Delivery records
- * Postpartum follow-up records

6. Antenatal Care

6.1 Recording a Visit

Step 1:

From the sidebar, click 'Antenatal Care'.

Step 2:

Select the patient from the list or search for them.

Step 3:

Click 'New Visit' to record a check-up.

Step 4:

Enter the visit details:

- * Date of visit
- * Gestational age (weeks)
- * Blood pressure, weight, fundal height
- * Fetal heart rate
- * Clinical notes and observations

Step 5:

Click 'Save Visit' to record the data.

6.2 Follow-up Schedule

The system helps track follow-up appointments. After each visit, you can schedule the next visit date. The dashboard shows upcoming follow-ups.

7. Investigations

7.1 Adding Lab Results

Step 1:

From the sidebar, click 'Investigations'.

Step 2:

Select the patient.

Step 3:

Click 'Add Investigation'.

Step 4:

Select the investigation type (e.g., Complete Blood Count, Blood Glucose, Urinalysis, Ultrasound).

Step 5:

Enter the results and any notes.

Step 6:

Click 'Save'.

7.2 Viewing Investigation History

All investigations for a patient are displayed in chronological order on their profile. You can view trends over time, compare results, and see which values are outside the normal range.

8. Classification & Risk Assessment

The system includes a machine learning-based classification engine that helps assess patient risk. It analyzes patient data and visit history to provide risk scores and recommendations.

Step 1:

From the sidebar, click 'Classification'.

Step 2:

Select a patient.

Step 3:

Review the automatically calculated risk factors.

Step 4:

The system displays risk level (Low, Moderate, High) and recommendations.

Classification considers: maternal age, BMI, blood pressure, blood glucose levels, previous pregnancy complications, and other clinical parameters.

9. Delivery Management

The delivery module tracks labor and delivery details for each patient.

Step 1:

From the sidebar, click 'Delivery'.

Step 2:

Select the patient.

Step 3:

Record delivery details:

- * Delivery date and time
- * Delivery mode (normal vaginal, C-section, vacuum, forceps)
- * Duration of labor
- * Complications (if any)
- * Baby details: weight, length, head circumference, Apgar scores

Step 4:

Click 'Save Delivery Record'.

Delivery records are stored in the patient's history for future reference.

10. Postpartum Care

The postpartum module tracks follow-up care for both mother and baby after delivery.

Step 1:

From the sidebar, click 'Postpartum'.

Step 2:

Select the patient.

Step 3:

Record postpartum visit details:

- * Date of follow-up
- * Mother: blood pressure, temperature, uterine involution, lochia, wound healing
- * Baby: feeding status, weight, jaundice check, vaccination status

Step 4:

Click 'Save'.

Regular postpartum follow-up is recommended at 1 week, 2 weeks, and 6 weeks after delivery.

11. Admin Panel

The Admin panel is accessible to users with the Admin role.

11.1 User Management

Administrators can manage user accounts:

- * View all registered users
- * Approve or reject new user registrations
- * Edit user roles (Admin, Doctor, Nurse, Receptionist)
- * Deactivate or delete user accounts

Step 1:

From the sidebar, click 'Admin' then 'User Management'.

Step 2:

Pending users appear with an 'Approve' button.

Step 3:

Click 'Approve' to grant access, or 'Reject' to deny.

11.2 Facility Management & Branding

Each facility can customize its branding. This controls what patients see when they log in:

- * Clinic Name: Displayed in the login box, sidebar, and footer
- * Support Phone: Displayed in the login box and footer
- * Clinic Logo: Upload a custom image (JPG/PNG) displayed on the login page and sidebar

Step 1:

Go to Admin -> Facility Management -> Clinic Branding.

Step 2:

Enter the clinic name and support phone number.

Step 3:

Click 'Choose File' to upload a logo image.

Step 4:

Click 'Save Branding' to update.

Changes take effect immediately. Each facility sees only its own branding.

12. Troubleshooting

Application does not start

- * Ensure you have Python 3.8+ installed (if running from source)
- * Check that port 5000 is not in use by another application
- * On Windows, run start.bat as Administrator
- * On Linux/macOS, ensure the start.sh file has execute permissions (chmod +x start.sh)

Browser does not open automatically

Manually open your browser and go to <http://localhost:5000/dashboard>

Cannot log in

- * Check that Caps Lock is off
- * Use the default credentials: admin / admin123
- * If you registered a new account, ask an administrator to approve it
- * Verify the facility code is correct

Language not switching properly

Click the language toggle button again. If the issue persists, refresh the page (F5).

13. Support

For technical support, questions, or feedback:

* Developer: Karim Abdelaziz

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We welcome your feedback to help improve the system.